

Computer Organization And Assembly Language

```
00000000: 01001101 01011010 10010000 00000000 00000011 00000000 MZ....
00000006: 00000000 00000000 00000100 00000000 00000000 00000000 .....
0000000c: 11111111 11111111 00000000 00000000 10111000 00000000 .....
00000012: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000018: 01000000 00000000 00000000 00000000 00000000 00000000 @....
0000001e: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000024: 00000000 00000000 00000000 00000000 00000000 00000000 .....
0000002a: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000030: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000036: 00000000 00000000 00000000 00000000 00000000 00000000 .....
0000003c: 10000000 00000000 00000000 00000000 00001110 00011111 .....
00000042: 10111010 00001110 00000000 10110100 00001001 11001101 .....
00000048: 00100001 10111000 00000001 01001100 11001101 00100001 !..L.!
0000004e: 01010100 01101000 01101001 01110011 00100000 01110000 This p
00000054: 01110010 01101111 01100111 01110010 01100001 01101101 rogram
0000005a: 00100000 01100011 01100001 01101110 01101110 01101111 canno
00000060: 01110100 00100000 01100010 01100101 00100000 01110010 t be r
00000066: 01110101 01101110 00100000 01101001 01101110 00100000 un in
0000006c: 01000100 01001111 01010011 00100000 01101101 01101111 DOS mo
00000072: 01100100 01100101 00101110 00001101 00001101 00001010 de....
00000078: 00100100 00000000 00000000 00000000 00000000 00000000 $.....
0000007e: 00000000 00000000 01010000 01000101 00000000 00000000 ..PE..
```

Lab Manual 07

Objectives:

1. To learn how shift and rotate instructions work
2. To Print values of registers using shift/rotate and logic instructions
3. To understand and apply how can we use shift instructions to multiply and divide with power of 2.

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LAB TASK

NOTE: You are required to prepare the ASM file of the following task. Use maximum number of procedures to do the tasks.

NOTE: Read instructions carefully.

- Use hard coded values to evaluate functions.
- Use only shift instructions for multiplication and division purposes (You can also use add/sub(tract) instructions where required).
- The procedures should accept first argument in bx/bl and second in dx/di(if applicable).
- Use logic instructions for the first two procedures.
- Use properly push and pop to avoid losing values of registers.

Write an assembly program which contains

- A procedure which accepts a number and checks whether it is even or odd.
- A procedure which accepts a character(ascii) and checks whether it is capital or small.
- A procedure which accepts a number as parameter and multiplies it by 8.
- A procedure which accepts a number as parameter and multiplies it by 16.
- A procedure which accepts a number as parameter and multiplies it by 2.
- A procedure which accepts a number as parameter and multiplies it by 4.
- A procedure which accepts a number as parameter and divide it by 8.
- A procedure which accepts a number as parameter and divide it by 16.
- A procedure which accepts a number as parameter and divide it by 2.
- A procedure which accepts a number as parameter and divide it by 4.
- A procedure which accepts a number and a power and multiplies the given number by given power of 2.
- A procedure which accepts a number and a power and divides the given number by given power of 2.
- A procedure which accepts a 16 bit number and prints it in binary.
- A procedure which accepts a 16 bit number and prints it in hex.
- A procedure which accepts a 8 bit number and prints it in binary.
- A procedure which accepts a 8 bit number and prints it in hex.
- A procedure which accepts a number and multiplies it by 10.
- A procedure which accepts a number and multiplies it by 13.
- A main procedure which will pass arguments to the procedures and show their implementation.